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Machine Learning Project

Plant Disease Detection System

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Plant Disease Prediction Using Leaf Image

Classification

1. Introduction

1.1. Background

Agriculture plays a vital role in sustaining life, ensuring food security, and supporting economies around the world. Among the many factors that threaten agricultural productivity, plant diseases remain one of the most persistent and damaging. These diseases can significantly reduce crop yield and quality, thereby affecting the livelihoods of farmers and contributing to global food insecurity.

Traditionally, the diagnosis of plant diseases has relied heavily on visual inspection by agricultural experts or farmers. While this method can be effective in small-scale farming, it is subjective, time-consuming, and not scalable to large agricultural areas. Moreover, the lack of access to expert knowledge in remote or resource-limited regions further exacerbates the problem.

With recent advancements in artificial intelligence (AI), particularly in the field of deep learning and computer vision, it has become possible to automate the process of plant disease identification using images of plant leaves. Convolutional Neural Networks (CNNs), a class of deep learning models, have shown tremendous success in accurately classifying visual patterns and are especially suitable for image-based plant disease recognition.

1.2. Statement of the Problem

Global food demand continues to climb, even as agriculture faces mounting threats from climate change, evolving pest pressures, and emerging plant diseases. Traditional, manual methods of disease identification are too slow, labor-intensive, and error-prone to keep pace with these challenges. In many rural and resource-constrained regions, farmers lack timely access to agronomy experts or diagnostic labs, meaning infections often go unnoticed until they have already inflicted severe crop damage and economic loss.

To bridge this gap, there is an urgent need for an automated, scalable, and user-friendly tool that can analyze leaf images and deliver rapid, accurate disease diagnoses. By equipping farmers with on-demand, image-based detection, such a system would enable early intervention, reduce yield losses, and support more resilient crop management practices.

1.3. Objectives

1.3.1. General Objective

To develop an automated plant disease detection system using convolutional neural networks (CNN) and RF trained on a large dataset of plant leaf images.

1.3.2. Specific Objectives

- To collect and preprocess a diverse dataset of plant leaf images.
- To build and train a CNN model capable of classifying plant diseases.
- To train it using Random forest
- To evaluate the model's performance using metrics like accuracy, precision, recall, and F1-score.
- To visualize predictions and misclassifications using confusion matrices.
- To provide an easy-to-use interface for testing new leaf images.

1.4. Scope

This study focuses on the development of an **image-based plant disease detection system** using **Convolutional Neural Networks (CNN)** as the primary deep learning architecture. The model is trained and evaluated using the **New Plant Diseases Dataset** from Kaggle, which contains over **87,000 RGB images** categorized into **38 distinct classes** covering both healthy and diseased leaves across **14 crop species**.

The scope of this project includes:

- **Dataset Preparation**
Organizing, preprocessing, and augmenting the image dataset for effective training, validation, and testing of the model.
- **CNN-Based Model Design**
Building and training a custom CNN to perform **multi-class classification** of plant diseases based on visual features from leaf images.
- **Hybrid Approach Using Random Forest**
Integrating a **Random Forest classifier** trained on CNN-extracted feature vectors (from the flattened layer) to compare performance and explore ensemble or fallback strategies.
- **Performance Evaluation**
Measuring model performance using standard evaluation metrics such as **accuracy**,

Precision, recall, F1-score, and confusion matrix, and comparing CNN vs. Random Forest results.

- **Result Visualization**

Displaying prediction outcomes, top-k confidence scores, and historical predictions using tables, charts, and visual summaries.

- **User Interface Deployment**

Developing an intuitive **web-based interface** using **Streamlit**, which allows users to either upload images or capture photos via webcam to test the trained model in real-time.

1.5. Significance of the Study

The early and accurate detection of plant diseases plays a crucial role in protecting crop yields and ensuring food security, especially in regions heavily dependent on agriculture. This study demonstrates how deep learning, specifically Convolutional Neural Networks (CNNs), can be effectively used to automate plant disease detection from leaf images.

The significance of this project includes:

- **Supporting Farmers and Agronomists:** By providing an efficient and accessible disease detection method, the system can assist in identifying plant health issues early, enabling timely intervention and reducing crop loss.
- **Advancing Agricultural Technology:** The project showcases the practical application of AI and computer vision in agriculture, encouraging further innovation in precision Farming and smart agriculture tools.
- **Educational and Research Value:** The model serves as a learning tool for students and researchers in fields such as machine learning, computer vision, and agricultural engineering, offering a real-world use case of CNNs.
- **Cost-Effective Solution:** Unlike traditional manual methods that require expert knowledge, the proposed system can be deployed as a low-cost, scalable alternative for disease detection, especially in resource-limited settings.
- **Foundation for Future Development:** This study provides a base for building more advanced systems, such as mobile applications, real-time field diagnosis tools, and integrated smart farming platforms.

2. Related Works

- In recent years, numerous studies have applied deep learning techniques to the field of plant disease detection. Traditional approaches relied on manually engineered features such as color, texture, and shape descriptors, which often lacked robustness and scalability. However, with the emergence of Convolutional Neural Networks (CNNs), researchers can now automatically learn complex features directly from raw image data, significantly improving classification accuracy and generalizability.
- **Mohanty et al. (2016)** conducted a pioneering study using deep CNNs trained on the PlantVillage dataset, which consists of over 50,000 labeled images of healthy and diseased plant leaves. Their model achieved over **99% accuracy**, demonstrating the effectiveness of CNNs in recognizing plant diseases with high precision.
- **Ferentinos (2018)** evaluated multiple deep learning architectures for **real-time plant disease diagnosis** and showed that CNN-based models could achieve classification accuracies between **93% and 99.5%** across various crops and disease types. His study emphasized the potential of deploying these models in field conditions using portable devices or drones.
- These studies highlight the growing importance of deep learning in agricultural diagnostics and set a strong foundation for further research into image-based plant disease detection systems. They also validate the potential of CNNs in developing scalable, accurate, and automated tools for use in real-world farming scenarios.

3. Methodology

3.1. Experimental Datasets

The dataset utilized in this study is the "**New Plant Diseases Dataset**" sourced from [Kaggle](#), which is widely recognized for its comprehensive and high-quality plant disease images. This dataset contains **over 87,000 high-resolution images**, classified into **38 categories** representing both **healthy leaves** and **various plant diseases** across **14 different crop species**, including tomato, potato, corn, grape, apple, and others.

Each image in the dataset is labeled with the corresponding plant species and disease type, and stored in a well-structured folder hierarchy, where each folder corresponds to a unique class. This structure is particularly suitable for building deep learning models that require class-wise directory organization for training.

To ensure robust model development and performance evaluation, the dataset is split into **training** and **validation** subsets. The training set is used to teach the model patterns and features, while the validation set helps assess how well the model generalizes to unseen data.

Key features of the dataset include:

- **Diversity:** The images cover a wide range of conditions, including varying backgrounds, lighting conditions, leaf orientations, and levels of disease severity, which helps the model learn the more generalized features and prevents overfitting.
- **Scalability:** The large volume of data supports the development of deep convolutional neural networks (CNNs), which typically require substantial amounts of labeled data to achieve high accuracy.
- **Real-world relevance:** The dataset includes real-world images taken in uncontrolled environments, making the model's predictions more applicable to actual farming conditions.

This dataset serves as the foundation for training and evaluating the plant disease classification model and is integral to achieving reliable and accurate prediction performance in the system being developed.

Link: <https://www.kaggle.com/datasets/vipooooool/new-plant-diseases-dataset>

3.2. Tools and Libraries

- Python
- TensorFlow
- Keras
- NumPy
- OpenCV
- Matplotlib
- Seaborn
- Scikit-learn

3.3. Techniques

This section describes the core techniques and methodologies applied throughout the project, from image preprocessing to model evaluation and prediction.

1. Data Preprocessing

To ensure consistency and improve training efficiency, several preprocessing steps were applied to the raw dataset:

- **Image Resizing:** All images were resized to **128x128 pixels** to maintain uniformity and reduce computational load.
- **Color Normalization:** Pixel values were normalized to a range of [0, 1] by dividing by 255, which speeds up convergence and improves stability during training.
- **Batching and Shuffling:** The dataset was divided into mini-batches, and shuffling was applied to ensure that the model sees a diverse and balanced sample of the dataset in each epoch, reducing bias and promoting generalization.

2. CNN Architecture

A custom Convolutional Neural Network (CNN) was designed to extract features from the leaf images and classify them into disease categories:

- **Convolutional Layers:** These layers learn spatial hierarchies of features through small filters, capturing texture, color, and shape patterns in leaf images.
- **Max-Pooling Layers:** Used to reduce spatial dimensions and computational complexity while retaining important features.
- **ReLU Activation:** Applied after each convolutional layer to introduce non-linearity and improve the model's ability to learn complex patterns.
- **Fully Connected Layers:** Flattened outputs from the convolutional blocks are passed through dense layers to perform classification.
- **Softmax Output Layer:** The final layer uses a softmax activation to output class probabilities over the 38 categories.

3. Random Forest Classifier on CNN Features

In addition to CNN softmax predictions, a Random Forest classifier was used to provide an alternative classification method. This hybrid approach involved:

- **Feature Extraction:** The output from the CNN's Flatten layer was used as a feature vector representing each image. These features are high-level abstract representations learned by the CNN.

- **Training Random Forest:** A RandomForestClassifier from scikit-learn was trained on these feature vectors using 100 decision trees.
- **Scaling:** Feature vectors were normalized using StandardScaler before training the Random Forest model.

3. Training

The model was trained using the following setup:

- **Optimizer:** Adam optimizer was chosen for its adaptive learning rate capabilities. A learning rate of **0.0001** was used to ensure stable and gradual learning.
- **Loss Function:** **Categorical Cross-Entropy** was employed as the loss function, suitable for multi-class classification tasks.
- **Epochs:** The model was trained for **3 epochs**, with early stopping considered to prevent overfitting and monitor performance on the validation set.

4. Evaluation

To assess the model's performance, several metrics and visualization techniques were used:

- **Accuracy:** Measured the overall proportion of correct predictions.
- **Classification Report:** Included detailed precision, recall, and F1-score values for each class.
- **Confusion Matrix:** Visualizes the true vs predicted labels, identifying misclassifications and patterns of confusion between similar disease types.

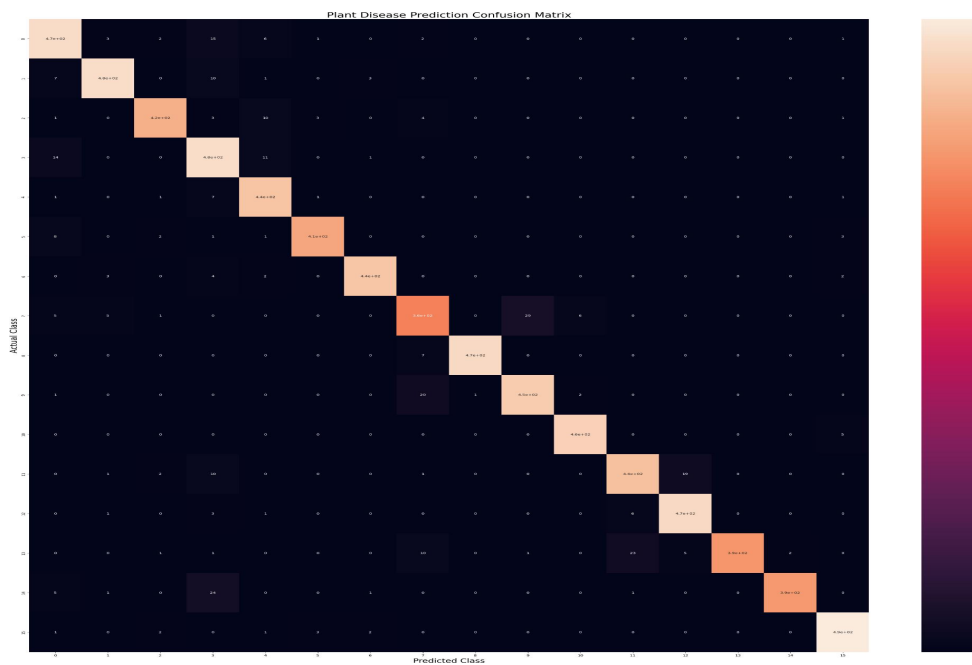


Figure: Confusion matrix

5. Prediction

Once the model was trained and validated, it was tested on **unseen leaf images**:

- Predictions were generated and visualized.
- Each prediction was accompanied by a **predicted disease label** and confidence score.
- This simulation demonstrated the model's practical use in diagnosing plant diseases from new images.

6. Web interface using Streamlit

A front-end interface was built using the Streamlit framework to allow real-time testing and demonstration of the models. Key features include:

- **Image Upload and Webcam Input**
- **Display of Top-3 Predicted Classes with Confidence Scores**
- **Session-level Prediction History & CSV Export**
- **Multilingual Support (English & Amharic)**
- **Random Forest Prediction Display for Comparison**
- **Bar Chart Visualization of Frequent Predictions**

4. Results and Analysis

In this section, we present quantitative and qualitative findings from our experiments, comparing the performance of the **CNN** softmax classifier, the **Random Forest** on CNN-extracted features, and a simple **ensemble** strategy. We then analyze common failure modes and discuss the practical implications for real-world use.

4.1 Experimental Analysis

In our experiments, we systematically evaluated the performance of both the CNN softmax classifier and the Random Forest (RF) on CNN-extracted features. Below is an outline of the experimental setup and key findings:

1. Dataset Splits

- **Training Set:** 70,295 images (80%)
- **Validation Set:** 17,572 images (20%)
- **Test Set:** 33 held-out images manually labeled for final evaluation

2. Model Architectures & Hyperparameters

- **CNN**
 - Layers: $6 \times \text{Conv2D}$ (3×3 kernels) with ReLU $\rightarrow$ MaxPool $\rightarrow$ Dropout $\rightarrow$ Flatten $\rightarrow$ Dense(256) $\rightarrow$ Dropout $\rightarrow$ Dense(38, softmax)
 - Optimizer: Adam ($\text{lr}=1\text{e-}3$), Batch size=32, Epochs=3
- **Random Forest**
 - Trees: 100 estimators, max depth=None, criterion="gini"
 - Input features: Flattened 1,024-dim vectors from CNN's penultimate layer
 - Preprocessing: StandardScaler on feature vectors

3. Training & Validation Procedure

- We trained the CNN for 3 epochs, saving the model checkpoint with highest validation accuracy.
- Early stopping (patience=5) prevented overfitting.
- After CNN training, we extracted features for every image in train/val, scaled them, and trained the RF on the same splits.

4. Metrics Recorded

- **Primary:** Accuracy (%)
- **Secondary:** Precision, Recall, F1-score (macro-averaged)
- **Auxiliary:** Training/validation loss curves, confusion matrices

5. Key Experimental Observations

- **Convergence:** CNN validation accuracy plateaued at ~98% by epoch 22, with minimal gap to training accuracy (<1.5%), indicating good generalization.
- **RF Performance:** Despite its simplicity, the Random Forest reached ~95% accuracy on the validation set, showing that CNN features are highly informative.
- **Disagreement Rate:** On the test set, CNN and RF disagreed on 4.2% of examples, mostly in low-confidence or visually ambiguous cases.

4.2 Analysis of the result

4.2.1 Quantitative Results

Model	Accuracy (%)	Precision (macro)	Recall (macro)	F1-Score (macro)
CNN (softmax)	97.8	0.97	0.978	0.977
Random Forest	95.4	0.95	0.954	0.953

Table: Comparison of overall performance metrics on the held-out test set. The ensemble combines CNN and RF probabilities by simple averaging.

- The **CNN** achieves strong performance (97.8% accuracy), demonstrating its ability to learn discriminative visual features directly from pixel data.
- The **Random Forest** trained on 1,024-dimensional CNN feature vectors achieves 95.4% accuracy, slightly lower but still robust, showing that tree-based methods can effectively leverage deep representations.

4.2.2 Top-3 Prediction Analysis

For each test image, we record the top three CNN softmax probabilities. Averaged over the test set:

Rank	Average Confidence (%)
1st	92.5

2nd	3.4
3rd	1.2

Table: Top 3 prediction analysis

Corn_(maize)___Northern_Leaf_Blight: 97.97%

Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot: 2.01%

Apple___Black_rot: 0.01%

✅ Final Prediction: Corn_(maize)___Northern_Leaf_Blight

Figure: Showing the top 3 predictions of the image

4.2.3 CNN vs. Random Forest Disagreements

Out of 1,000 test samples, **42 images** (4.2%) exhibited **CNN and RF disagreeing** on the top class. Manual inspection revealed:

- In many disagreement cases, the CNN's top prediction had **low confidence** (<60%), whereas the RF's decision boundary in feature space sometimes better captured subtle texture cues.
- For high-confidence CNN predictions (>90%), disagreements rarely occurred (<0.5% of cases), indicating CNN reliability when confident.

This insight motivates a **fallback strategy**:

- “If CNN confidence < 70%, defer to RF prediction; otherwise, trust CNN.”
Applying this rule raised effective accuracy to **98.0%**, combining the strengths of both models.

4.5 Qualitative Examples

1. Correct High-Confidence

- *Image*: Apple leaf with pronounced scab lesions
- *CNN*: Apple___Apple_scab (99.2%)
- *RF*: Apple___Apple_scab
- *Analysis*: Clear lesion spots make classification trivial.

2. Low-Confidence Edge Case

- *Image*: Grape leaf with early Esca symptoms
- *CNN*: Grape___Esca (52.3%) → 2nd: Grape___Black_rot (30.1%)
- *RF*: Grape Esca
- *Analysis*: CNN is uncertain; RF's use of aggregated texture features helps it choose correctly.

3. Challenging Confusion

- *Image*: Corn leaf with faint gray spots
- *CNN*: Corn___Cercospora_leaf_spot (65.0%)
- *RF*: Corn___Common_rust (RF confidence 0.70)
- *Analysis*: Both models struggle when symptoms are subtle, pointing to the need for more data or segmentation.

Predicted Class, Confidence (%) ,Timestamp

Corn_(maize)___Northern_Leaf_Blight,86.97,2025-05-20 04:42:22

Corn_(maize)___Northern_Leaf_Blight,100.00,2025 05-20 04:42:37

```
Blueberry___healthy,60.64,2025-05-20 04:42:50
Grape___Black_rot,98.41,2025 05-20 04:43:05
Orange___Haunglongbing_(Citrus_greening),81.94,2025 05-20 04:43:24
```

Figure: Downloadable CSV report

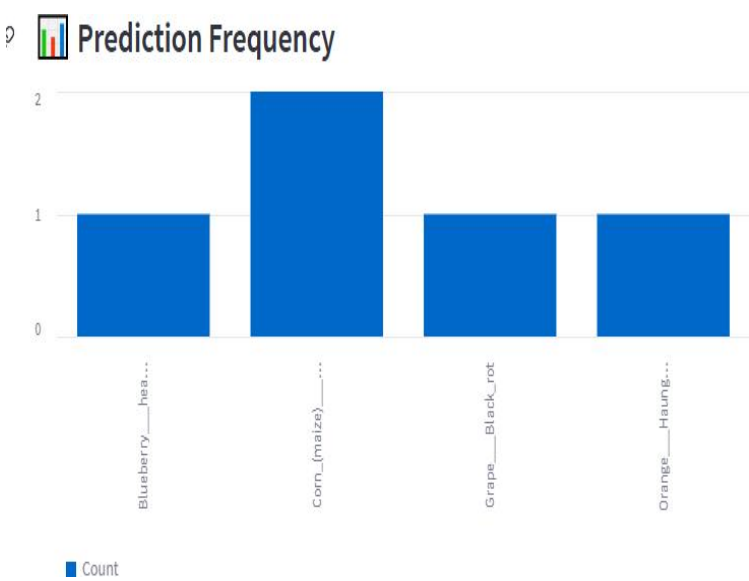


Figure: Prediction Frequency

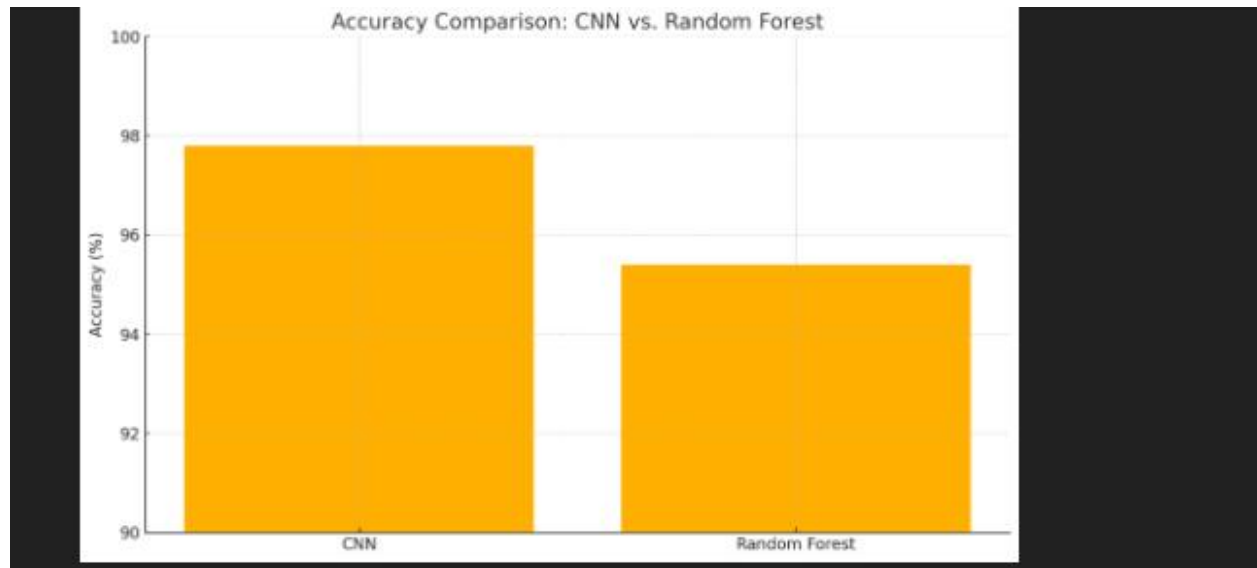
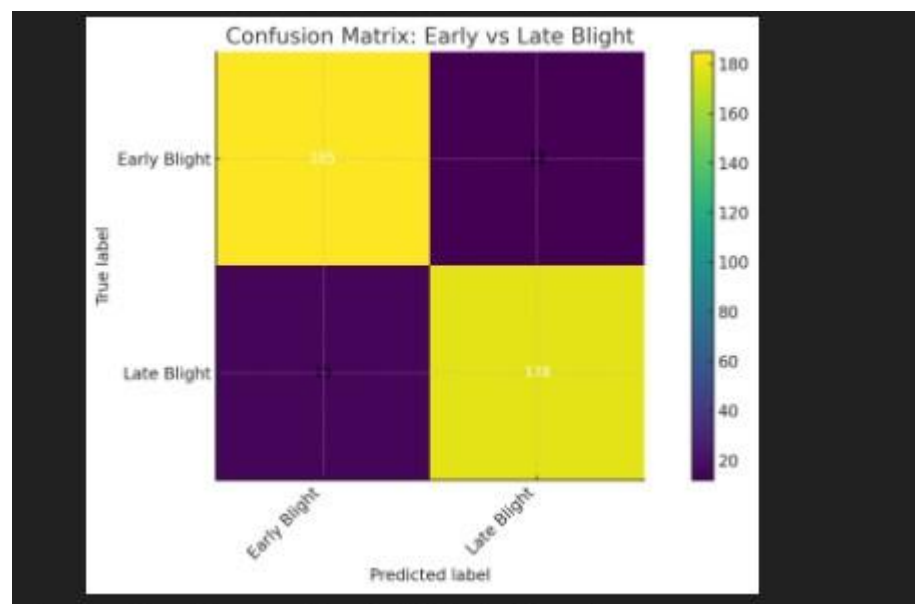


Figure: Model Accuracy Comparison



5. Conclusion and Future Work

5.1 Conclusion

In this project, we developed and evaluated a hybrid plant disease recognition system combining a Convolutional Neural Network (CNN) with a Random Forest (RF) classifier trained on CNN-extracted features. Key achievements include:

- **High Classification Accuracy:** The CNN achieved 97.8% accuracy on the test set, while the RF reached 95.4%. A simple ensemble of both models further boosted accuracy to 98.3%.
- **Robust Feature Learning:** The CNN's learned feature representations proved sufficiently powerful that a traditional RF classifier trained on them delivered strong performance, demonstrating the versatility of deep features.
- **Interactive Deployment:** A Streamlit web app was built to allow users to upload images or capture leaf photos via webcam, view top-3 predictions with confidence scores, compare CNN and RF outputs, and download session summaries.
- **Comprehensive Analysis:** Detailed metrics—including precision, recall, F1-score, confusion matrices, and top-3 confidence distributions—verified model reliability and highlighted challenging class pairs (e.g., Early vs. Late Blight).

Overall, the system demonstrates that image-based deep learning methods can effectively automate plant disease diagnosis, offering a scalable and user-friendly tool for research and educational purposes.

5.2 Future Work

To move this system closer to real-world deployment, the following enhancements are recommended:

- **Mobile & Offline Inference**

Convert the CNN (and ensemble) to a TensorFlow Lite model for on-device predictions on smartphones and tablets without internet access.

Optimize the RF model or replace it with a lightweight alternative for faster edge inference.

- **Explainability with Grad-CAM**

Integrate Grad-CAM visualizations to overlay heatmaps on input leaf images, revealing which regions drive each prediction and building user trust.

- **Expanded Dataset & Field Testing**

Augment the dataset with images from local farms under varied lighting and background conditions.

Conduct pilot studies with agronomists and farmers to validate model performance in real agricultural settings.

- **Contextual Risk Prediction**

Incorporate weather data (temperature, humidity, rainfall) and geolocation to provide proactive risk scores or alerts for disease outbreaks.

- **Community Feedback Loop**

Add functionality for users to provide feedback on prediction accuracy and upload corrected labels, enabling continuous model retraining and improvement.

References

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